

Example submission Readme:

output of mvn --version

```
C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06>mvn --version
Apache Maven 3.8.1 (05c21c65bdfed0f71a2f2ada8b84da59348c4c5d)
Maven home: C:\Program Files\apache-maven-3.8.1\bin\..
Java version: 16.0.1, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-16.0.1
Default locale: en_CA, platform encoding: Cp1252
OS name: "windows 10", version: "10.0", arch: "amd64", family: "windows"
```

Output of mvn compile

```
C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp>mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< SEG3103:BookstoreApp >-----
[INFO] Building BookstoreApp 0.1.0
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ BookstoreApp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\src\main\resources
[INFO]
[INFO] --- maven-compiler-plugin:3.7.0:compile (default-compile) @ BookstoreApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.747 s
[INFO] Finished at: 2021-07-08T17:31:20-04:00
[INFO]
[INFO] -----
```

output of mvn package -DskipTests

```
C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp>mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< SEG3103:BookstoreApp >-----
[INFO] Building BookstoreApp 0.1.0
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ BookstoreApp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\src\main\resources
[INFO]
[INFO] --- maven-compiler-plugin:3.7.0:compile (default-compile) @ BookstoreApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.747 s
[INFO] Finished at: 2021-07-08T17:31:20-04:00
[INFO]
[INFO] -----

C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp>mvn package -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] -----< SEG3103:BookstoreApp >-----
[INFO] Building BookstoreApp 0.1.0
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ BookstoreApp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\src\main\resources
[INFO]
[INFO] --- maven-compiler-plugin:3.7.0:compile (default-compile) @ BookstoreApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ BookstoreApp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\src\test\resources
[INFO]
[INFO] --- maven-compiler-plugin:3.7.0:testCompile (default-testCompile) @ BookstoreApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.1:test (default-test) @ BookstoreApp ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- maven-jar-plugin:2.4:jar (default-jar) @ BookstoreApp ---
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.747 s
[INFO] Finished at: 2021-07-08T17:31:20-04:00
[INFO]
[INFO] -----
```

```
[INFO] META-INF/LICENSE already added, skipping
[WARNING] Configuration options: 'appendAssemblyId' is set to false, and 'classifier' is missing.
Instead of attaching the assembly file: C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\target\BookstoreApp-0.1.0.jar, it will become the file for main project artifact.
NOTE: If multiple descriptors or descriptor-formats are provided for this project, the value of this file will be non-deterministic!
[WARNING] Replacing pre-existing project main-artifact file: C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\target\BookstoreApp-0.1.0.jar
with assembly file: C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\target\BookstoreApp-0.1.0.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.496 s
[INFO] Finished at: 2021-07-08T17:33:45-04:00
[INFO] -----
```

This command will tell maven to compile the project, take the compiled project and package it into a jar file which is stored in the target directory. using `-DskipTests` tells maven that you do not want to execute the tests when it is done packaging, so it will skip them.

```
C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp>java -jar ./target/BookstoreApp-0.1.0.jar
Press Enter to stop server
```

output of java -jar bookstore5.jar

```

C:\Users\Rikar\OneDrive\Desktop\SchoolFiles\2021SummerSEGI_playground\Labs\BookstoreApp>java -jar bookstore.jar
Spring Boot : (v2.1.4.RELEASE)

2021-07-08 17:42:53.549 INFO 11304 --- [main] org.springframework.boot.SpringApplication : Starting BookstoreApplication v0.0.1-SNAPSHOT on rick-honek with PID 11304 (C:\Users\Rikar\OneDrive\Desktop\SchoolFiles\2021SummerSEGI\2021SummerSEGI_Labs\BookstoreApp\bookstore.jar)
2021-07-08 17:42:53.600 INFO 11304 --- [main] org.springframework.boot.SpringApplication : No active profile set, falling back to default profiles: default
2021-07-08 17:42:54.008 INFO 11304 --- [main] s.c.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data repositories in DEFAULT mode.
2021-07-08 17:42:54.029 INFO 11304 --- [main] s.c.d.r.c.RepositoryConfigurationDelegate : Found 1 repository interfaces.
2021-07-08 17:42:54.429 INFO 11304 --- [main] org.springframework.transaction.annotation.ProxyTransactionManagementConfiguration : Bean 'org.springframework.transaction.annotation.ProxyTransactionManagementConfiguration' of type [org.springframework.transaction.annot
ation.ProxyTransactionManagementConfiguration$HibernateProxy] cannot be eligible for
Tomcat initialization with port(s) 8080 (http
gettin
g processed by all beans/processors (for example: not eligible for auto-proxying))
2021-07-08 17:42:54.745 INFO 11304 --- [main] o.a.p.s.catalina.core.StandardService : Starting service [Tomcat].
2021-07-08 17:42:54.756 INFO 11304 --- [main] o.a.p.s.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.17]
2021-07-08 17:42:54.816 INFO 11304 --- [main] o.a.w.c.c.CatContext{localhost}[/] : Initializing Spring embedded WebApplicationContext
2021-07-08 17:42:54.880 INFO 11304 --- [main] org.springframework.context.support.GenericWebApplicationContext : Initialization completed in 1224 ms
2021-07-08 17:42:55.018 INFO 11304 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 Starting...
2021-07-08 17:42:55.059 INFO 11304 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 Start completed.
2021-07-08 17:42:55.118 INFO 11304 --- [main] o.hibernate.jpa.internal.util.LogHelper : HHH000043: Processing PersistenceUnitInfo [
name: default
, #000000412: Hibernate Core (5.3.9.Final)
, #000000413: Hibernate properties (no
HCANON000001: Hibernate Commons Annotations (5.6.4.Final)
, #000000048: Using dialect: org.hibernate.dialect.H2Dialect
, #000000049: Creating schema: internal.exec.SchemaInputOutputInstantImp[5e92c21']
Initializ
ed JPA EntityManagerFactory for persistence unit 'default'
User access (securityCode=0) is considered unsafe for production and is only used for sample applications.
, #000000050: Concurrent ThreadPooledTaskExecutor
, #000000051: Initializing ExecutorService 'applicationTaskExecutor'
, #000000052: spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure
spring.jpa.open-in-view to disable this warning
, #000000053:
main() s.o.s.web.DefaultServletInitializerChain : Creating filter chain: any request; [org.springframework.security.web.servlet.config.SecurityWebMvcIntegrationFilter#web,o, org.springframewor
k.security.web.authentication.UsernamePasswordAuthenticationFilter#csrfFilter, org.springframework.security.web.header.HeaderWriterFilter#csrfHeaderWriter, org.springframework.security.web.authentication.InputOutputFilter#addReferer, org.springf
ramework.security.web.authentication.AnonymousAuthenticationFilter#anonymousFilter, org.springframework.security.web.access.ExceptionTranslationFilter#exceptionTranslator, org.springframework.security.web.access.Exception
TranslationFilter#exceptionTranslator];
, #000000054: main() s.o.s.web.DefaultServletInitializerChain : Tomcat started on port(s): 8080 (http) with context path [/]
, #000000055: main() s.o.s.web.DefaultServletInitializerChain : Started BookstoreApplication in 3.861s
```

Output of mvn test

```
C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp>mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< SEG3103:BookstoreApp >-----
[INFO] Building BookstoreApp 0.1.0
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ BookstoreApp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\src\main\resources
[INFO]
[INFO] --- maven-compiler-plugin:3.7.0:compile (default-compile) @ BookstoreApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ BookstoreApp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\src\test\resources
[INFO]
[INFO] --- maven-compiler-plugin:3.7.0:testCompile (default-testCompile) @ BookstoreApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.1:test (default-test) @ BookstoreApp ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running ExampleTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.018 s - in ExampleTest
[INFO] Running selenium.ExampleSeleniumTest
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Starting ChromeDriver 91.0.4472.101 (af52a90bf87030dd1523486a1cd3ae25c5d76c9b-refs/branch-heads/4472@#1462)) on port 7325
Only local connections are allowed.
Please see https://chromedriver.chromium.org/security-considerations for suggestions on keeping ChromeDriver safe.
ChromeDriver was started successfully.
Jul. 08, 2021 5:44:05 P.M. org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
Starting ChromeDriver 91.0.4472.101 (af52a90bf87030dd1523486a1cd3ae25c5d76c9b-refs/branch-heads/4472@#1462)) on port 14389
Only local connections are allowed.
Please see https://chromedriver.chromium.org/security-considerations for suggestions on keeping ChromeDriver safe.
ChromeDriver was started successfully.
Jul. 08, 2021 5:44:07 P.M. org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.166 s - in selenium.ExampleSeleniumTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.980 s
[INFO] Finished at: 2021-07-08T17:44:08-04:00
[INFO] -----
```

Screenshot of at least one additional selenium web driver test

```
83  @Test
84  public void f3() {
85      driver.findElement(By.id("searchBtn")).click();
86      WebElement book = driver.findElement(By.id("order-hall001"));
87      book.click();
88      driver.findElement(By.id("cartLink")).click();
89
90      boolean isFound = false;
91      int numRows = driver.findElements(By.xpath("//div[@class='content']/table/tbody/tr")).size(); //find number of rows in case more than 1 book is added
92
93      for(int i=1; i<=numRows; i++){
94          //find by xpath starting with the content div
95          if(driver.findElement(By.xpath("//div[@class='content']/table/tbody/tr["+i+"]/td[1]")).getText().equals("hall001")){
96              isFound=true;
97          }
98      }
99      assertEquals(true, isFound);
100  }
```

Output of mvn test for the new test

```
C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp>mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< SEG3103:BookstoreApp >-----
[INFO] Building BookstoreApp 0.1.0
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ BookstoreApp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\src\main\resources
[INFO]
[INFO] --- maven-compiler-plugin:3.7.0:compile (default-compile) @ BookstoreApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ BookstoreApp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\src\test\resources
[INFO]
[INFO] --- maven-compiler-plugin:3.7.0:testCompile (default-testCompile) @ BookstoreApp ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 3 source files to C:\Users\Ricar\OneDrive\Desktop\SchoolFiles\2021Summer\SEG3103_playground\Lab06\BookstoreApp\target\test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.1:test (default-test) @ BookstoreApp ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running ExampleTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.024 s - in ExampleTest
[INFO] Running selenium.ExampleSeleniumTest
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Starting ChromeDriver 91.0.4472.101 (af52a90bf87030dd1523486a1cd3ae25c5d76c9b-refs/branch-heads/4472@{#1462}) on port 12269
Only local connections are allowed.
Please see https://chromedriver.chromium.org/security-considerations for suggestions on keeping ChromeDriver safe.
ChromeDriver was started successfully.
Jul. 08, 2021 7:54:35 P.M. org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
Starting ChromeDriver 91.0.4472.101 (af52a90bf87030dd1523486a1cd3ae25c5d76c9b-refs/branch-heads/4472@{#1462}) on port 21944
Only local connections are allowed.
Please see https://chromedriver.chromium.org/security-considerations for suggestions on keeping ChromeDriver safe.
ChromeDriver was started successfully.
Jul. 08, 2021 7:54:37 P.M. org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
Starting ChromeDriver 91.0.4472.101 (af52a90bf87030dd1523486a1cd3ae25c5d76c9b-refs/branch-heads/4472@{#1462}) on port 27115
Only local connections are allowed.
Please see https://chromedriver.chromium.org/security-considerations for suggestions on keeping ChromeDriver safe.
ChromeDriver was started successfully.
Jul. 08, 2021 7:54:38 P.M. org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.432 s - in selenium.ExampleSeleniumTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 8.711 s
[INFO] Finished at: 2021-07-08T19:54:39-04:00
[INFO]
[INFO] -----
```